

ASSIGNMENT 3

DUCTING LAYOUT AND PRESSURE-DROP CALCULATION

Industrial dust-extraction design based on the supplied plant drawings

DESIGN OUTPUT

Three representative duct systems | Darcy-Colebrook method

Critical-path pressure loss | Branch balancing | Fan and motor duty

All unverified dimensions and coefficients are explicitly declared assumptions.

Name: _____

Roll No.: _____

Course / Instructor: _____

Submission date: _____

Design summary and governing results

Scope statement

The supplied drawings are low-resolution plant layouts. They establish equipment order and approximate topology, but many dimensions and labels are unreadable. This report therefore presents a transparent preliminary design using stated industrial assumptions. It is suitable as an academic assignment and design basis; a field survey is required before fabrication or purchase.

System	Duty	Flow	Critical path	Req. FTP	Selected FTP	Motor
A	Conveyor extraction	10000 m ³ /h	A3	3402 Pa	3750 Pa	18.5 kW
B	Multi-level extraction	15000 m ³ /h	B1	3380 Pa	3750 Pa	30.0 kW
C	Cooler and bin extraction	44000 m ³ /h	C2	4690 Pa	5200 Pa	110.0 kW

Main conclusions

System A is governed by branch A3. The calculated fan total pressure is 3402 Pa, so 3750 Pa is selected. The calculated shaft power is 16.03 kW and the next practical motor size is 18.5 kW.

System B is governed by branch B1. The calculated fan total pressure is 3380 Pa, so 3750 Pa is selected. The calculated shaft power is 24.04 kW and a 30 kW motor is selected.

System C is governed by the new-cooler branch C2. The calculated fan total pressure is 4690 Pa, so 5200 Pa is selected. The calculated shaft power is 97.78 kW and a 110 kW motor is selected.

Interpretation

FTP means fan total pressure at the design flow. FSP is estimated as FTP minus the stack outlet velocity pressure. Final selection must use the supplier fan curve at the stated air density and include sound, erosion, dust loading, and motor service requirements.

Drawing interpretation and solution plan

What was extracted from the drawings

Drawing	Readable process intent	Representative model adopted
1	Conveyor hoods, collecting duct, bag collector, fan and stack	Four equal pickups in two submains; 10,000 m ³ /h
2	Multi-level pickups and risers feeding collector, fan and stack	Four elevated pickups in two submains; 15,000 m ³ /h
3	Existing/new cooler and bin zones, collector/fan and old-fan/stack	Six equivalent zones; existing/new submains; 44,000 m ³ /h

Calculation sequence

- 1 Define simultaneous hood flows and assumed straight lengths.
- 2 Select circular duct sizes to keep dusty-air velocity at or above 18 m/s.
- 3 Calculate velocity, Reynolds number, Darcy friction factor and velocity pressure.
- 4 Add straight-duct friction and local fitting losses section by section.
- 5 Sum each complete source-to-stack route and identify the largest total.
- 6 Balance all shorter routes to the critical route at their design flows.
- 7 Add 10% pressure margin, round upward, and calculate fan and motor duty.

Boundary of confidence

No blurred drawing dimension is treated as verified. Duct lengths, diameters, hood flows, bend counts, junction coefficients and collector loss are calculation assumptions. Each is shown so the design can be updated directly after field measurement.

Design basis and declared assumptions

Item	Adopted value	Use in calculation
Air density	1.20 kg/m ³	Velocity pressure and Reynolds number
Dynamic viscosity	1.81 x 10 ⁻⁵ Pa.s	Reynolds number
Steel roughness	0.15 mm	Colebrook friction factor
Dusty transport velocity	>= 18 m/s	Branch, submain and dirty main sizing
Collector allowance	1200 Pa	Base differential pressure
Fan total efficiency	65%	Shaft power
Motor efficiency	90%	Input power
Operation	All pickups simultaneous	No diversity or leakage allowance
Duct geometry	Circular internal diameter	Area and D in equations
Old-fan tie-in in C	Closed/isolated	No reverse or parallel flow

Preliminary fitting coefficients

Component	K	Reference velocity / boundary
Simple hood entry (A and B)	0.50	Own branch velocity
Equivalent collection zone (C)	2.00	Own branch velocity; includes unresolved pickup resistance
Long-radius 90-degree bend	0.25 each	Velocity in its duct section
Incoming combining junction	0.50	Velocity of the incoming section
External gradual adapter	0.15	Velocity of assigned section
Stack terminal including exit	1.50	Stack velocity

Coefficient status

These K values are declared preliminary design assumptions. Exact coefficients depend on elbow radius, area ratio, branch angle, vane construction and the reference velocity used by the selected fitting table.

Pressure-drop method

Flow and velocity

$$Q = Q_h / 3600 \quad A = \pi D^2 / 4 \quad V = Q / A$$

$$Re = \rho V D / \mu \quad p_v = \rho V^2 / 2$$

Darcy-Colebrook friction

$$1/\sqrt{f} = -2 \log_{10} [\epsilon / (3.7D) + 2.51 / (Re \sqrt{f})]$$

The Colebrook equation was iterated until the numerical residual was below 1×10^{-10} . The Darcy factor is then used directly; no Fanning-factor conversion is required.

Section loss

$$\Delta p_f = f (L/D) p_v \quad \Delta p_K = K p_v$$

$$\Delta p_{\text{section}} = [f(L/D) + \sum K] p_v$$

Critical route and fan

$$\Delta p_{\text{route}} = \text{sum(duct and fitting losses)} + \Delta p_{\text{collector}}$$

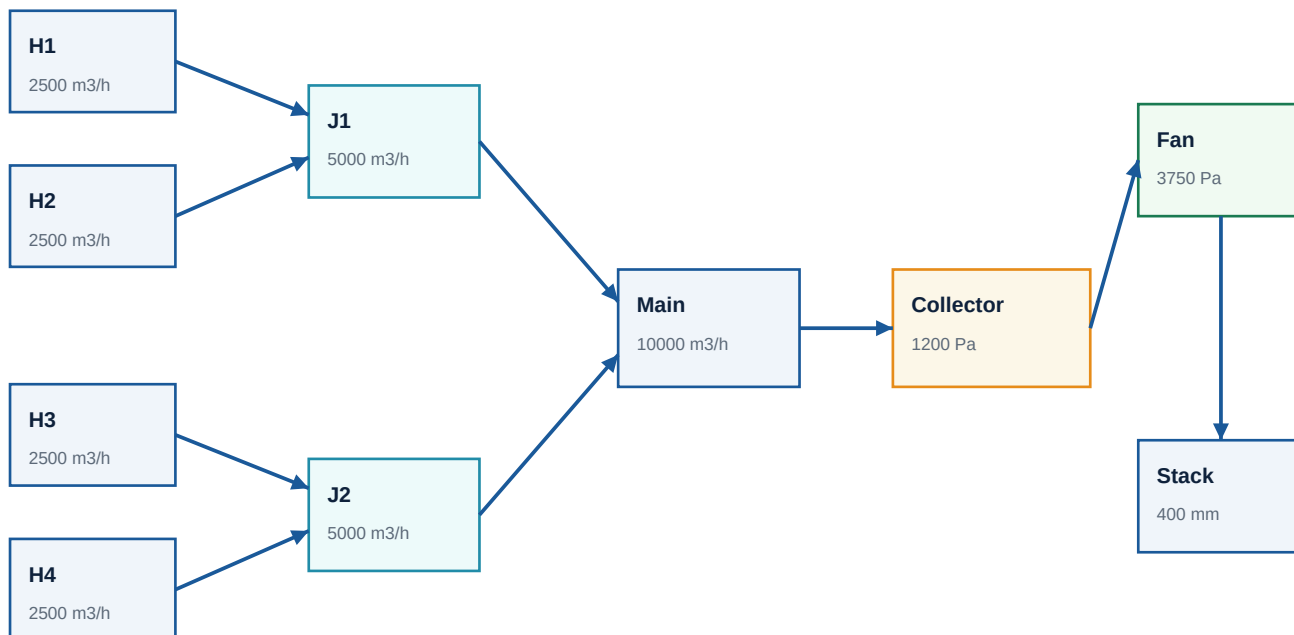
Required FTP = max(Δp_{route}). Selected FTP = the required value increased by 10% and rounded upward to the next 50 Pa.

$$P_{\text{shaft}} = Q \times \text{FTP} / \eta_{\text{fan}} \quad P_{\text{input}} = P_{\text{shaft}} / \eta_{\text{motor}}$$

Network rule

Losses in series are added. Parallel branch losses are not added together. At the design flows, each active branch route is balanced to the same source-to-discharge pressure difference.

System A: Conveyor extraction



Critical route shown by the highest calculated total; branch balancing is added separately.

Assumed geometry and calculated section performance

ID	Qh	D	L	V	Re/1e3	f	K	dp_f	dp_K	dp
A1	2500	200	8.0	22.10	293	0.0195	1.50	229.0	439.8	668.7
A2	2500	200	6.0	22.10	293	0.0195	1.25	171.7	366.5	538.2
A3	2500	200	10.0	22.10	293	0.0195	1.75	286.2	513.1	799.2
A4	2500	200	5.0	22.10	293	0.0195	1.25	143.1	366.5	509.6
AS1	5000	300	12.0	19.65	391	0.0179	0.75	165.9	173.7	339.6
AS2	5000	300	6.0	19.65	391	0.0179	0.75	82.9	173.7	256.7
AM	10000	400	10.0	22.10	586	0.0167	0.65	122.3	190.6	312.8
ACL	10000	450	4.0	17.47	521	0.0165	0.40	26.8	73.2	100.0
AE	10000	400	12.0	22.10	586	0.0167	2.00	146.7	586.3	733.1

Units: Qh in m3/h; D in mm; L in m; V in m/s; pressure losses in Pa.

Sizing check

Dusty sections range from NaN to NaN m/s. The minimum exceeds the adopted 18 m/s transport criterion.

System A: route totals and fan duty

Complete source-to-stack routes

Source	Route	Route total	Balance dp	Added K
A1	A1-AS1-AM-FILTER-ACL-AE	3354.2	47.6	0.16
A2	A2-AS1-AM-FILTER-ACL-AE	3223.7	178.1	0.61
A3	A3-AS2-AM-FILTER-ACL-AE	3401.8	0.0	0.00
A4	A4-AS2-AM-FILTER-ACL-AE	3112.2	289.7	0.99

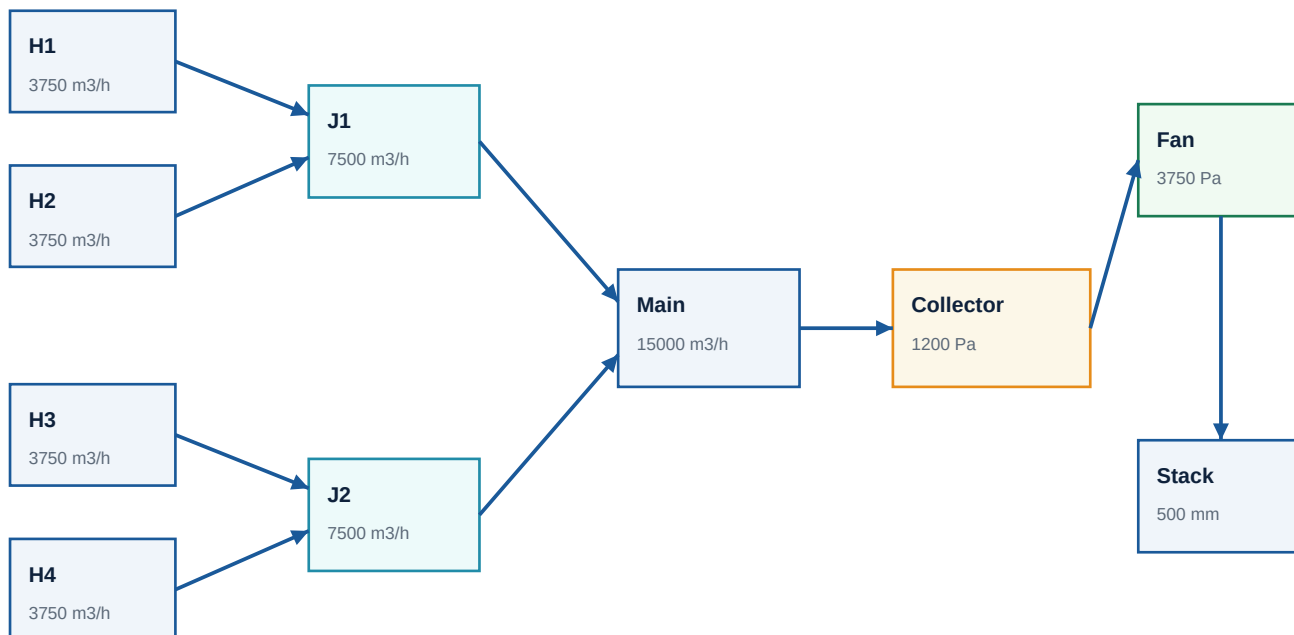
Critical route

A3 -> AS2 -> AM -> FILTER -> ACL -> AE gives 3401.8 Pa. Other branches require the balancing losses listed above so all prescribed flows can coexist.

Fan selection

Quantity	Calculated / selected value
System volume flow	2.778 m ³ /s (10000 m ³ /h)
Required fan total pressure	3401.8 Pa
Selected fan total pressure	3750 Pa (10% allowance, rounded upward)
Estimated fan static pressure	3456.8 Pa
Fan shaft power	16.03 kW at 65% total efficiency
Electrical input	17.81 kW at 90% motor efficiency
Recommended motor rating	18.5 kW

System B: Multi-level extraction



Critical route shown by the highest calculated total; branch balancing is added separately.

Assumed geometry and calculated section performance

ID	Qh	D	L	V	Re/1e3	f	K	dp_f	dp_K	dp
B1	3750	250	16.0	21.22	352	0.0186	1.75	321.2	472.8	794.1
B2	3750	250	12.0	21.22	352	0.0186	1.50	240.9	405.3	646.2
B3	3750	250	10.0	21.22	352	0.0186	1.50	200.8	405.3	606.0
B4	3750	250	8.0	21.22	352	0.0186	1.25	160.6	337.7	498.3
BS1	7500	350	8.0	21.65	502	0.0172	1.00	110.6	281.3	391.9
BS2	7500	350	6.0	21.65	502	0.0172	0.75	82.9	211.0	293.9
BM	15000	500	8.0	21.22	703	0.0159	0.65	68.9	175.6	244.5
BCL	15000	560	5.0	16.92	628	0.0158	0.40	24.2	68.7	92.9
BE	15000	500	13.5	21.22	703	0.0159	2.00	116.2	540.4	656.6

Units: Qh in m³/h; D in mm; L in m; V in m/s; pressure losses in Pa.

Sizing check

Dusty sections range from NaN to NaN m/s. The minimum exceeds the adopted 18 m/s transport criterion.

System B: route totals and fan duty

Complete source-to-stack routes

Source	Route	Route total	Balance dp	Added K
B1	B1-BS1-BM-FILTER-BCL-BE	3379.9	0.0	0.00
B2	B2-BS1-BM-FILTER-BCL-BE	3232.1	147.9	0.55
B3	B3-BS2-BM-FILTER-BCL-BE	3094.0	286.0	1.06
B4	B4-BS2-BM-FILTER-BCL-BE	2986.3	393.7	1.46

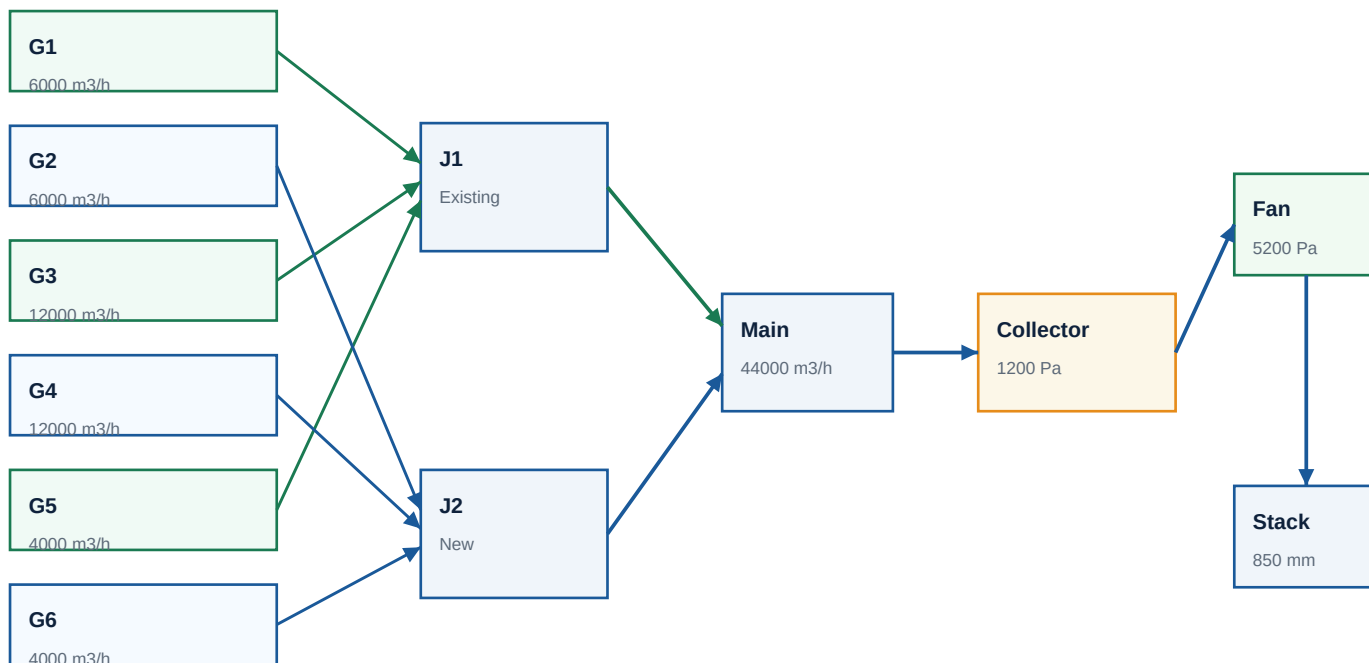
Critical route

B1 -> BS1 -> BM -> FILTER -> BCL -> BE gives 3379.9 Pa. Other branches require the balancing losses listed above so all prescribed flows can coexist.

Fan selection

Quantity	Calculated / selected value
System volume flow	4.167 m ³ /s (15000 m ³ /h)
Required fan total pressure	3379.9 Pa
Selected fan total pressure	3750 Pa (10% allowance, rounded upward)
Estimated fan static pressure	3479.8 Pa
Fan shaft power	24.04 kW at 65% total efficiency
Electrical input	26.71 kW at 90% motor efficiency
Recommended motor rating	30.0 kW

System C: Cooler and bin extraction



Assumed geometry and calculated section performance

ID	Qh	D	L	V	Re/1e3	f	K	dp_f	dp_K	dp
C1	6000	300	38.0	23.58	469	0.0177	3.25	748.7	1084.1	1832.8
C2	6000	300	44.0	23.58	469	0.0177	3.50	866.9	1167.5	2034.4
C3	12000	450	22.0	20.96	625	0.0163	3.25	210.1	856.6	1066.7
C4	12000	450	28.0	20.96	625	0.0163	3.50	267.4	922.5	1189.9
C5	4000	250	18.0	22.64	375	0.0185	3.00	409.7	922.2	1332.0
C6	4000	250	24.0	22.64	375	0.0185	3.25	546.3	999.1	1545.4
CS1	22000	625	20.0	19.92	825	0.0153	1.00	116.2	238.1	354.3
CS2	22000	625	16.0	19.92	825	0.0153	1.00	93.0	238.1	331.0
CM	44000	850	40.0	21.54	1214	0.0143	0.65	186.8	180.9	367.7
CCL	44000	900	8.0	19.21	1146	0.0142	0.40	27.9	88.6	116.5
CE	44000	850	18.0	21.54	1214	0.0143	2.00	84.1	556.7	640.8

Units: Qh in m3/h; D in mm; L in m; V in m/s; pressure losses in Pa.

Sizing check

Dusty sections range from NaN to NaN m/s. The minimum exceeds the adopted 18 m/s transport criterion.

System C: route totals and fan duty

Complete source-to-stack routes

Source	Route	Route total	Balance dp	Added K
C1	C1-CS1-CM-FILTER-CCL-CE	4512.1	178.4	0.53
C2	C2-CS2-CM-FILTER-CCL-CE	4690.4	0.0	0.00
C3	C3-CS1-CM-FILTER-CCL-CE	3746.0	944.5	3.58
C4	C4-CS2-CM-FILTER-CCL-CE	3845.9	844.5	3.20
C5	C5-CS1-CM-FILTER-CCL-CE	4011.2	679.2	2.21
C6	C6-CS2-CM-FILTER-CCL-CE	4201.4	489.0	1.59

Critical route

C2 -> CS2 -> CM -> FILTER -> CCL -> CE gives 4690.4 Pa. Other branches require the balancing losses listed above so all prescribed flows can coexist.

Fan selection

Quantity	Calculated / selected value
System volume flow	12.222 m ³ /s (44000 m ³ /h)
Required fan total pressure	4690.4 Pa
Selected fan total pressure	5200 Pa (10% allowance, rounded upward)
Estimated fan static pressure	4921.6 Pa
Fan shaft power	97.78 kW at 65% total efficiency
Electrical input	108.64 kW at 90% motor efficiency
Recommended motor rating	110.0 kW

Fully worked example: System A branch A3

Branch A3 is shown in full so the result can be reproduced. Values are retained during calculation and rounded only for presentation.

1. Flow, area and velocity

$$Q = 2500/3600 = 0.694444 \text{ m}^3/\text{s}$$

$$A = \pi(0.200)^2/4 = 0.03141593 \text{ m}^2$$

$$V = Q/A = 22.105 \text{ m/s}$$

2. Reynolds number and velocity pressure

$$Re = (1.20)(22.104853)(0.200)/(1.81 \times 10^{-5}) = 293103$$

$$p_v = (1.20)(22.104853)^2/2 = 293.175 \text{ Pa}$$

3. Friction and fitting loss

$$\epsilon/D = 0.00015/0.200 = 0.000750; \text{ Colebrook gives } f = 0.01952358.$$

$$\Delta p_f = 0.01952358(10/0.200)(293.175) = 286.19 \text{ Pa}$$

$$K = 0.50 \text{ hood} + 0.50 \text{ junction} + 3(0.25 \text{ bends}) = 1.75$$

$$\Delta p_K = 1.75(293.175) = 513.06 \text{ Pa}$$

$$\Delta p_{A3} = 286.19 + 513.06 = 799.25 \text{ Pa}$$

4. Complete critical path

Component	Pressure loss
A3	799.25 Pa
AS2	256.66 Pa
AM	312.83 Pa
FILTER	1200.00 Pa
ACL	100.03 Pa
AE	733.07 Pa

$$\text{Total} = 799.25 + 256.66 + 312.83 + 1200.00 + 100.03 + 733.07 = 3401.83 \text{ Pa.}$$

Critical-path loss breakdown

System A

Component	Loss (Pa)	Share
A3	799.2	23.5%
AS2	256.7	7.5%
AM	312.8	9.2%
FILTER	1200.0	35.3%
ACL	100.0	2.9%
AE	733.1	21.5%

System B

Component	Loss (Pa)	Share
B1	794.1	23.5%
BS1	391.9	11.6%
BM	244.5	7.2%
FILTER	1200.0	35.5%
BCL	92.9	2.7%
BE	656.6	19.4%

System C

Component	Loss (Pa)	Share
C2	2034.4	43.4%
CS2	331.0	7.1%
CM	367.7	7.8%
FILTER	1200.0	25.6%
CCL	116.5	2.5%
CE	640.8	13.7%

Filter-loading sensitivity and design checks

Sensitivity to collector differential pressure

System	Collector dp (Pa)	Required FTP (Pa)	Rounded FTP with 10% (Pa)
A	800	3002	3350
A	1200	3402	3750
A	1600	3802	4200
B	800	2980	3300
B	1200	3380	3750
B	1600	3780	4200
C	800	4290	4750
C	1200	4690	5200
C	1600	5090	5600

The base selection uses 1200 Pa. A dirtier collector raises the system resistance nearly one-for-one. Fan control and cleaning strategy must therefore be checked against the expected operating differential-pressure band.

Engineering checks

OK	All dusty sections meet the adopted minimum transport velocity of 18 m/s.
OK	All calculated flows are conserved at the combining junctions.
OK	The critical path is the largest prescribed-flow route in each system.
OK	Balancing K values are based on each branch velocity pressure, matching the stated reference boundary.
OK	A positive pressure allowance exists above the base calculated resistance.
OK	Fan static pressure is reported separately from fan total pressure.
OK	Motor selection is based on shaft power and rounded to a practical nominal rating.

Branch balancing schedule

The damper duty is the additional pressure needed to make each non-critical route equal to the critical route at the stated flow. The K values are preliminary targets for balancing-device selection and commissioning.

System	Branch	Flow m3/h	Unbalanced Pa	Add Pa	Target K
A	A1	2500	3354.2	47.6	0.16
A	A2	2500	3223.7	178.1	0.61
A	A3	2500	3401.8	0.0	0.00
A	A4	2500	3112.2	289.7	0.99
B	B1	3750	3379.9	0.0	0.00
B	B2	3750	3232.1	147.9	0.55
B	B3	3750	3094.0	286.0	1.06
B	B4	3750	2986.3	393.7	1.46
C	C1	6000	4512.1	178.4	0.53
C	C2	6000	4690.4	0.0	0.00
C	C3	12000	3746.0	944.5	3.58
C	C4	12000	3845.9	844.5	3.20
C	C5	4000	4011.2	679.2	2.21
C	C6	4000	4201.4	489.0	1.59

Commissioning note

Set dampers by measured branch flow, not solely by blade angle. After the collector reaches its normal operating condition, traverse accessible straight runs, adjust the lowest-resistance branches, and repeat until the scheduled flows are obtained.

Conclusion, limitations and references

Conclusion

The three interpreted networks were sized and evaluated with the Darcy-Weisbach equation, iterated Colebrook friction factors, and explicit local-loss coefficients. At the assumed simultaneous flows, Systems A, B and C require 3402 Pa, 3380 Pa and 4690 Pa of fan total pressure, respectively. The recommended preliminary selections are 3750 Pa with an 18.5 kW motor for A, 3750 Pa with a 30 kW motor for B, and 5200 Pa with a 110 kW motor for C.

The pressure totals are internally consistent with the declared topology, lengths and coefficients. The largest remaining uncertainty is the unreadable information in the scans, especially physical lengths, fitting geometry, hood resistance and collector operating differential pressure. These inputs should be replaced by site measurements and vendor data before fabrication.

Final verification required

Confirm every route length, internal diameter, branch angle, elbow radius, hood type and equipment pressure loss. Obtain the collector clean/normal/dirty pressure band. Then rerun the same section schedule and select a fan whose stable operating point lies on the supplier curve at the actual air density.

References

1. NIOSH. Handbook for Dust Control in Mining. DHHS (NIOSH) Publication.
2. McGill AirFlow. Design Advisory 01: Airflow, friction and pressure fundamentals.
3. McGill AirFlow. Design Advisory 02: System effect, critical path and balancing.
4. ASHRAE Utah Chapter / SMACNA. Duct Design Fundamentals.
5. Figueroa et al. Hood entry-loss relationship for local exhaust systems.
6. Donaldson. Differential pressure in dust collectors.

Submission note

This document intentionally separates observed drawing intent from assumed numerical inputs. That distinction is essential to a defensible engineering calculation.